Personal Statement

Fulbright–Masaryk Postdoctoral Fellowship

Jan Saro, Ph.D., MBA | Czech University of Life Sciences Prague & Institute of Information Theory and Automation, Czech Academy of Sciences

An engineer who chose agriculture

I was trained as a computer engineer at the Czech Technical University, and I spent the first decade of my career writing software for banks, payment networks, and public-sector information systems — EESSI for European social-security exchange, citizen-facing tools at the Ministry, security and privacy services at Avast. The work was useful, but at some point I noticed that the people who most needed good software — farmers, veterinarians, family-run businesses outside Prague — were the people the technology industry was building for last. That observation moved me out of commercial software and into a Ph.D. in Systems Engineering at the Czech University of Life Sciences, which I defended in 2025. Research is the part of this trajectory I am most enthusiastic about: I now hold a postdoctoral track in which I get to study, full time, how artificial intelligence and operations research can become a quiet, dependable assistant for a dairy farmer at five in the morning — and I would rather do that than anything else.

The Fulbright–Masaryk programme appeals to me precisely because it asks two questions at once: *is your science good enough?* and *will the public be better off because of it?* I have spent the last four years building my answer to both.

What I do, and why dairy

My doctoral thesis built data-driven decision-support systems and predictive models for dairy herd health: Markov chains for disease transitions, a herd-health DSS published in the *Czech Journal of Animal Science*, and disease-scoring tools that farmers can use without a data scientist in the room. The work earned the Dean's Award for the best scientific publication at FEM CZU in 2024 and, more importantly, has been picked up by people who actually run cows. In the same year, I was selected as a finalist for the Falling Walls Lab Czech Republic for a talk titled *Breaking the Wall of the Dairy Farmer AI Assistant* — a deliberate attempt to translate the work into language a non-specialist audience could follow. I now hold parallel appointments as Assistant Professor at CULS and Junior Researcher at the Institute of Information Theory and Automation of the Czech Academy of Sciences, which gives me one foot in agricultural reality and the other in mathematical rigour.

Dairy matters in Central Europe for reasons that are not always visible from the United States. It is one of the few remaining sectors where small and medium family farms still compete with large operations; it carries a disproportionate share of the EU's agricultural methane footprint, which I have begun to study with colleagues; and it sits at the intersection of food security, animal welfare and rural livelihood. A decision-support tool that genuinely respects the farmer's expertise is therefore not only a technical artefact — it is a small instrument of fairness in a sector where the gap between data-rich and data-poor producers is widening fast.

Civic engagement: the line I will not cross

The civic engagement core of this proposal is a commitment I have already made in writing on the Czech side, and that I will keep on the U.S. side: every line of code, every dataset transformation, and every model produced under the Fulbright–Masaryk fellowship will be *open by default*, *auditable by a non-specialist*, and *welfare-first by design.* There is real temptation in agricultural AI to optimise yield at the expense of the animal, or to lock decision logic behind proprietary platforms that small farms cannot afford. I have turned down freelance offers for both. The line is not difficult to hold; it just has to be held publicly.

This is also why I am applying to a postdoctoral fellowship rather than to industry. The right venue for a tool that small farms and extension veterinarians can use is a university with a public-good mandate, working alongside the U.S. extension tradition. The Fulbright–Masaryk programme aligns those incentives in a way that no commercial post would.

Why the United States, why now

I have studied dairy decision-support across the U.S., European, and Australasian literature carefully. The U.S. research community has produced the field's most influential herd-level simulators and decision-support frameworks; the U.S. extension system is, despite its pressures, still the closest working example of academia delivering everyday tools to working farmers. I want to spend four months inside that culture, not as a visitor reading the papers, but as a researcher contributing to one of the host group's active workstreams. Four months is also the right size of commitment for a young academic with teaching obligations at home — enough to build, validate and publish a hybrid digital twin prototype; short enough that my Ph.D. students in Prague do not lose continuity.

What I will bring back

I do not view a Fulbright stay as something I receive — I view it as a shared output that should leave both sides materially better off. The stay is built around two concrete artefacts that benefit the host institution and the Czech research base equally. The first is a working digital-twin prototype, hardened to a level at which a company or a start-up — in the United States, in the Czech Republic, or jointly — could adopt it as the basis of a deployment, in the spirit of a grant whose output is a usable technology rather than a closed report. The second is a peer-reviewed scientific publication on the design and validation of the twin, with co-authorship for members of the host group secured from the outset of the stay. Both artefacts are vehicles for the broader purpose of the Fulbright–Masaryk programme: to widen and deepen scientific cooperation between the Czech Republic and the United States. The publication anchors a long-term research relationship, the prototype anchors a long-term applied-research relationship, and together they give both institutions something they can build on after I return home.

A short word about how I work

I am a builder by temperament. I prefer a working prototype over a clean slide. I have run a freelance web-development practice for over a decade alongside academic work — more than fifty delivered websites — and I know how to ship under real-world constraints. I have an MBA, which means I can talk to funders, vendors, and farm owners in their own language. I am at C1 in English. Applied AI is also a personal enthusiasm of mine, not only a professional one — I build small AI tools in my free time, follow the field closely outside working hours, and keep returning to the question of how to make these models genuinely useful to people who are not engineers. I am thirty-five, which is recent enough after my Ph.D. defence to qualify as a postdoctoral fellow but mature enough to know which projects are worth four months of my life. This one is.

Respectfully submitted,
Jan Saro, Ph.D., MBA